IT 6903 – Scientific Communication

Complete Model Answers · Batch 17 Past Paper (February 2026)

University of Moratuwa · M.Sc / PG Diploma in Artificial Intelligence
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Question 1 — Communication, Language & AI [25 marks]

(a) Human communication and AI [3 × 3 = 9 marks]

(i) Distinctive feature of human communication

Answer: The distinctive feature is **symbolic language** — the ability to use symbols such as words, letters, and numbers to manipulate and communicate meaning.

Unlike animal communication (signals, calls, body language), humans can invent and share **stories** and knowledge that are preserved across generations. Through symbolic language we can collect, store, transmit, and discover new knowledge. Lecture emphasis: humanity became so capable over other animals largely due to this communication ability (storytelling for the next generation). Symbolic manipulation of language also underpins mathematics, science, engineering, and computing.

(ii) Technologies for man–man, man–machine, machine–machine

(p) Man–man: telephone, email, video conferencing, written documents (papers/thesis), messaging apps.

(q) Man–machine: chatbots / LLMs (e.g. ChatGPT), computers, smartphones, expert systems, brain–computer interfaces (BCI), EEG-based systems.

(r) Machine–machine: ATM ↔ bank servers, APIs between services, Multi-Agent Systems (MAS), IoT device-to-device protocols. Lecture note: machine-to-machine communication is now possible without humans.

(iii) "AI is approaching the technological singularity in language processing."

Position: Agree **partially**, with strong caveats.

Modern LLMs (e.g. ChatGPT) show impressive language processing and can appear near-human in many tasks. However, singularity (machine intelligence surpassing and escaping human control in a transformative way) is **not yet reached** in language AI because of major limitations:

- **Hallucination** — models invent plausible but false content.
- **Explainability** — non-symbolic ML is hard to interpret; lecture stresses synergy with symbolic AI / neuro-symbolic approaches.
- Lack of reliable pragmatics, grounded understanding, and guaranteed truthfulness.

So AI is advancing fast in language processing, but claiming singularity is overstated until these gaps are closed.

(b) True / False with justification [4 × 3 = 12 marks]

(i) Biological programming enables communication between plants and/or animals.

TRUE. After Turing, people programmed computers; nowadays people also program plants and animals via **biological programming / biological computing**. Lecture: we are heading towards communication with plants and animals through biological computing — changing biological systems (including genetic-level programming) so they can be instructed/controlled in a communication-like way.

(ii) There is a synergy between recent developments in non-symbolic AI and symbolic AI in language processing.

TRUE. Non-symbolic AI (ML/LLMs/Transformers) is strong at generating language and learning from data, but weak at explainability. Symbolic AI (rules, logic, expert systems) provides explainability and structured reasoning. Lecture examples include addressing explainability in ML with symbolic AI / Expert Systems, and neuro-symbolic AI — combining both for better language systems.

(iii) Before the birth of AI, Alan Turing emphasized the role of communication in determining machine intelligence.

TRUE. The Turing Test judges intelligence by whether a machine can communicate so well that it fools a human into thinking it is human. Thus Turing placed communication at the centre of defining machine intelligence, before AI became an established field.

(iv) Syntax is the most critical concern when communicating algorithmically.

FALSE. A language has four major aspects — **lexicon, syntax, semantics, and pragmatics** — and **all four** must be developed. Coding shows this: wrong words (lexicon), wrong grammar (syntax), wrong meaning (semantics), or wrong order/flow (pragmatics) all break communication. Syntax alone is not the most critical concern.

(c) Cognitive features [4 marks]

(i) One education-related cognitive feature that enhances other abilities

Thinking (algorithmic / language-related thinking). Lecture: reading activates/wakes up thinking, leading to understanding; thinking supports memory formation (neural circuits). Improving thinking (via math, programming, AI exercises, games, puzzles) enhances related cognitive skills such as understanding, memory, and communication.

(ii) AI-based solution + one negative implication

Solution: An AI-based interactive educational game (or AI agent for attention/mindfulness training) that trains thinking through puzzles, problem-solving, and guided reflection.

Negative implication: gaming / technology addiction — overuse may reduce real-world learning, attention, or social engagement. (Lecture principle: technology should make life easier, but if removed we should not be paralyzed.)

Question 2 — Reading & Research Literacy [25 marks]

(a) Reading research papers [3 × 3 = 9 marks]

(i) Two broad categories of information + what a novice gains

Two broad categories extracted from research papers (lecture framing of scientific documents):

1. **Problem-related information** — background, gap, issues, failures/successes of others, importance of the problem.
2. **Solution-related information** — technologies, methods, approaches, design/implementation ideas, evaluations.

Additional gains for a novice: map of major universities/people/institutes/journals/conferences in the area; improved writing style and vocabulary; how arguments and citations are structured; references leading to further papers.

(ii) Three negative implications of AI-summarize → write proposal; analogy; correct use

Three immediate negative implications:

1. **Weak reading comprehension** — the student never deeply processes the papers.
2. **Weak analytical thinking** — cannot judge gaps, +/– of technologies, or define an original problem.
3. **Poor note-taking / ownership of knowledge** — no personal summaries; vocabulary and writing skill stagnate; risk of unread or hallucinated content entering the proposal.

Analogy: Always using GPS without learning the map — when the tool fails (or invents a wrong route), you are lost. Lecture: if you know how to write/read, technology improves you; otherwise technology kills your ability.

Correct use: (1) Human reads and does the cognitive task first (summaries in own words, notes, problem/technology extraction); (2) then ask the machine to improve language/structure — not to replace thinking.

(iii) Techniques that improve reading comprehension

(i) Reading aloud: doubles perceptual input (eyes + ears). Strengthens thinking and memory; lecture: reading a little loudly improves understanding.

(ii) Reading highlighted text: underlining/highlighting focuses attention on problem/technology concepts, supports remembrance and understanding, and guides later detailed reading of the body.

(iii) Automating left-to-right eye movement: the brain splits tasks — eye movement (right-brain skill) vs understanding (left-brain). If L→R scanning becomes automatic through practice, energy is freed for understanding rather than hunting words. Use a pen/ruler and avoid focusing beyond the current line.

(b) True / False with justification [4 × 3 = 12 marks]

(i) In double-blind review, reviewers send comments directly to authors.

FALSE. In the publication process, authors submit to the **editor**; reviewers assess quality and send comments to the editor; the editor decides and communicates with authors. Reviewers do not independently send comments directly to authors (especially in double-blind, identities are hidden).

(ii) Researcher credibility is assessed using multiple metrics derived from two main parameters.

TRUE. Credibility is commonly assessed via metrics built from **publications** and **citations** (and related indexes / impact factors). Examples: citation indexes (Scopus, Google Scholar, Web of Science), impact factor of journals, h-index-style measures derived from those bases.

(iii) Since knowledge is power, a bigger portion of finances in publishing goes to authors.

FALSE. Although authors create knowledge, the larger share of publishing income typically goes to **publishers** (and related commercial structures). Author royalties are comparatively small (lecture mentions royalty around ~10% in the publishing storyboard). Knowledge is power, but financial control of publication often sits with publishers/indexes.

(iv) Reading and writing are activities that naturally demand mindfulness.

TRUE. Many tasks (walking, eating, listening to music) can be done together; reading and writing generally cannot — they require attention on one thing so neural circuits can form. If a task is fully trained/automatic, mindfulness ≈ 0; reading/writing remain mindfulness-demanding cognitive work.

(c) Best practices in reading for a research project [4 marks]

(i) Best practices

- Read **Abstract** → **Introduction** → **Conclusion** first to identify problem and technology concepts.
- Highlight/underline important points; then read body for details of marked concepts.
- Write **summaries/notes in your own words** (problem, technology, +/–).
- Use a **reference manager** (e.g. Zotero) for download, tags, notes, related papers.
- Start from a review paper; follow its reference chain; organize reading around gap + technology.

(ii) Most influential practice energizing ≥3 cognitive skills

Note-taking / writing summaries in your own words is the most influential. A note acts as a **cognitive device**. It energizes at least:

1. **Thinking** — selecting and restructuring ideas;
2. **Understanding** — restating meaning forces comprehension;
3. **Memory** — writing helps remember what was read (lecture: writing helps remember what you wrote).

It also improves vocabulary and later writing of proposal/thesis chapters (Literature Review is a collection of such summaries).

Question 3 — Scientific Documents & Thesis Writing [25 marks]

(a) Scientific documents in the research process [3 × 3 = 9 marks]

(i) Four major scientific documents; content of the first; two topics

Four major documents: (1) **Research / Project Proposal**, (2) **Progress Report**, (3) **Conference Paper**, (4) **Thesis** (journal paper is a related later form for wider access).

Any scientific document talks about two main topics: **Problem** and **Solution** (course title framing: Problem + Solution).

First document = Project Proposal (Ability Report). Typical content:

Title (problem + technology), Introduction (gap + proposed solution), Objectives (problem-related + solution-related), Literature Review (others' work → problem/technology), Problem in Brief, Proposed Solution/Approach (hypothesis, input, output, process, features, users), Resources, Plan of Action, References.

Distribution of the two topics: Problem-heavy early (intro background, LR, problem definition); Solution-heavy later (approach/design note). Abstract often balances problem–solution–outcome.

(ii) Most influential chapter for developing the entire thesis

Literature Review. It is the turning point where you know the **problem/gap** and the **technology/inspiration**. From LR you can derive title, hypothesis, objectives, approach, and the direction of all later chapters (technology, design, implementation, evaluation). Without a strong LR, the thesis has no justified problem or method.

(iii) Stage to conceptualize the entire project; content to develop

Stage: after adequate reviewing, at the **Project Proposal** stage (when Approach can be conceptualized).

Content: hypothesis; input/output/process (technology); users; features; objectives; problem in brief; proposed solution; resources; plan of action mapped to objectives. Approach is the key step in the research process.

(b) True / False with justification [4 × 3 = 12 marks]

(i) Thesis chapters must be carefully developed in sequential chapter order.

FALSE. Writing is **incremental**, not strictly sequential. Practically, develop Literature Review and then Approach early; later expand Design, Implementation, Evaluation. Outline → draft → revise → fine-tune is bidirectional. You cannot write a thesis in one shot in strict chapter order from page 1.

(ii) In the Milestone Approach, "Methodology" is adapted to suit steps in computing projects.

TRUE. In computing/AI projects, the general term Methodology is operationalized as milestone stages such as Technology → Approach → Design → Implementation → Evaluation (rather than a vague single "methods" chapter). This suits software/AI research workflows and maps to proposal → progress → conference paper → thesis milestones.

(iii) For good reasons, both objectives and problem definition should come before the Literature Review.

FALSE. Objectives can appear early (proposal/intro structure), but **problem definition** should come **after** the Literature Review. The researcher must know the literature to define the gap properly. Putting problem definition before LR loses justification and confuses the reader.

(iv) Only a few Research Software offer facilities to develop all sections/chapters as per a research method.

TRUE. Few platforms support the full research-document lifecycle aligned to a method. Lecture points students to specialized support such as **InPRAWeb** (https://inpraweb.lib.uom.lk/) for developing research writing across sections/chapters — not general office tools alone.

(c) Outline–Draft–Revise–Fine-tune framework [4 marks]

(i) People involved in the four stages

- **Outline:** Student + Supervisor (agree structure/headings).
- **Draft:** mainly Student (write under headings; remove blank-page fear).

- **Revise:** Student (many times) + Supervisor (about 1–2 times) to improve content/flow.
- **Fine-tune:** language improvement — Student and/or language editor; supervisor may guide academic tone.

Justification: supervisor shapes research direction at outline/revise; student owns drafting; language polishing should not replace the student's thinking.

(ii) Using AI without harming language skills

Use AI **after** you have outlined and drafted in your own words. Ask AI to improve clarity, grammar, or structure of **your** draft — not to invent the research argument. Keep rewriting, note-taking, and reading active. Lecture rule: if you know how to write, technology improves you; otherwise it kills ability. Prefer: human cognitive work first → machine to refine.

Question 4 — Evaluation, Presentations & Human Skills [25 marks]

(a) Evaluations in the research process [3 × 3 = 9 marks]

(i) Major evaluation points and purpose of documents/presentations

1. **Proposal evaluation** — Proposal + presentation show ability to define gap/technology and conceptualize Approach; permission to proceed.
2. **Progress review** — Progress report + slides show advancement through Design/Implementation; evidence that objectives are being achieved.
3. **Thesis / final evaluation (viva)** — Thesis (+ often presentation) shows full research including Evaluation and Conclusion; candidate must defend ownership.

(Conference presentations are also formative evaluations for feedback before final defense.)

(ii) Two components of evaluations; final thesis evaluation methods

Two components: (1) **Written document** (proposal/report/thesis) and (2) **Verbal presentation / oral defense**.

Institutional variations for final evaluation:

- Viva only — strong test of ownership/communication; writing less weighted.
- Thesis only — strong on written scholarship; weaker check that student did the work.
- Viva + thesis (common) — balances document quality with defense; lecture notes cases where good thesis + bad viva can fail, and weak thesis + good viva may pass with corrections.

(iii) Difficult situations and possible examiner judgments

1. **Bad thesis + good viva** — may pass with minor/major corrections (oral shows understanding despite writing issues).
2. **Good thesis + bad viva** — may fail; examiners doubt ownership/confidence despite a strong document.
3. **Weak thesis but student manages Q&A** — examiners may still probe depth; outcomes depend on whether core contribution and honesty are clear.

Other difficulties: unclear objectives, weak evaluation design, inability to open relevant slides, overconfidence, or treating unachieved objectives as "further work."

(b) True / False with justification [4 × 3 = 12 marks]

(i) Both objectives and problem definition should appear after the Literature Review in any research presentation.

FALSE. In presentations (as in proposals), **Objectives** typically appear before/around early structure, while **problem definition** should follow Literature Review (gap justified by others' work). Putting both after LR — or putting problem before LR — loses the reader. Lecture progress/thesis slide order: Introduction → Objectives → Literature Review → Research Problem → Methodology…

(ii) Progress Review slides can be developed simply by adding research-process steps after the Approach.

FALSE. Progress Review needs substantive content for Design and Implementation (with diagrams), broader LR than the proposal, problem definition, discussion of ongoing work — not merely appending step names after Approach. Slides must match the progress document and show real progress.

(iii) During Q&A in a viva, it is inappropriate to use additional material such as notes beyond prepared slides.

TRUE. Lecture: do not look at additional notes — it shows you are not confident. Instead, open the relevant slide (if unsure, open the Approach slide). Listen fully, then answer.

(iv) Viva-voce examinations assess more than the technical knowledge of the candidate.

TRUE. Viva checks technical soundness **and** presentation skills, confidence/Q&A, project skills such as **getting on with others**, **facing criticism**, **reaching consensus**, and exposure to the research area (key people, institutes, journals, conferences).

(c) Biological limits vs machines; threatened human skill [4 marks]

(i) FIVE biological limitations of humans superseded by machines

1. **Processing speed** — machines compute far faster than biological brains for many tasks.
2. **Memory capacity / perfect recall** — machines store and retrieve vast data without forgetting.
3. **Fatigue / need for rest** — machines can run continuously; humans tire.
4. **Precision and consistency** — machines repeat tasks with lower error variance.
5. **Parallel multi-channel processing at scale** — machines handle massive parallel workloads (e.g. search, reading comprehension benchmarks) beyond human biological limits. Lecture also notes machines have outperformed humans in reading comprehension in some settings.

(ii) Human skill most threatened in the future

Language-related communication skills (especially routine reading, writing, customer-service dialogue, and some IT documentation/coding communication) are most threatened, because generative AI already automates large parts of language work. If humans outsource these skills entirely, comprehension and original expression decline — the course's core warning.

END OF MODEL ANSWERS — Revise with lecture slides; expand to mark length in the exam (justify T/F; use Problem + Solution framing throughout).